

Figure 16: FEA Stress Contour Maps (Pending Ground-Truth Re-Analysis)

(a) Von Mises Stress Contour — Original Geometry

IMAGE
PLACEHOLDER

FEA re-analysis required

Shows von Mises stress distribution
across the original house geometry
under ASCE 7-22 ASD loading.

Color map: Blue (low) → Red (high)
Threshold: $\sigma_{VM, allow} = 5.0$ MPa

Source: SfePy FEA solver output
Resolution: ~100K tetrahedral elements

(b) Von Mises Stress Contour — Optimized V11

IMAGE
PLACEHOLDER

FEA re-analysis required

Shows stress redistribution after
45% material removal. Critical for
validating surrogate predictions.

Expected: stress concentration at
wall-roof junctions and remaining
interior partition connections.

Status: FUTURE WORK